

Semantic Matching Engine

Day 12 Internship Deliverables

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Submitted To

Company: Zecpath

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1.Semantic matching engine

```
semantic_matching > app.py > ...
1  from matcher import semantic_similarity
2
3  with open("sample_resume.txt", "r", encoding="utf-8") as file:
4      resume = file.read()
5
6  with open("sample_job.txt", "r", encoding="utf-8") as file:
7      job = file.read()
8
9  score = semantic_similarity(resume, job)
10
11 print("\n==== SEMANTIC MATCHING ENGINE =====\n")
12 print(f"Resume ↔ JD Similarity : {score}%")
13
14 if score >= 80:
15     print("Match Status      : Excellent Match")
16 elif score >= 60:
17     print("Match Status      : Good Match")
18 elif score >= 40:
19     print("Match Status      : Average Match")
20 else:
21     print("Match Status      : Low Match")
```

2.Resume ↔ JD similarity scorer

```
semantic_matching > matcher.py > semantic_similarity
1  from sentence_transformers import SentenceTransformer
2  from sklearn.metrics.pairwise import cosine_similarity
3
4  # Load embedding model
5  model = SentenceTransformer("all-MiniLM-L6-v2")
6
7
8  def semantic_similarity(resume_text, job_text):
9      """
10     Returns semantic similarity score between
11     resume and job description.
12     """
13
14     embeddings = model.encode([resume_text, job_text])
15
16     score = cosine_similarity(
17         [embeddings[0]],
18         [embeddings[1]]
19     )[0][0]
20
21     return round(score * 100, 2)
```

3. Matching accuracy report

```
tokenizer_config.json: 100%|
vocab.txt: 100%|
tokenizer.json: 100%|
special_tokens_map.json: 100%|
config.json: 100%|

===== SEMANTIC MATCHING ENGINE =====

Resume ↔ JD Similarity : 89.80999755859375%
Match Status           : Excellent Match
(.venv) PS C:\Users\91949\OneDrive\Downloads\Desktop\my works\Zecpath-AI-System\semantic_matching>
```

```
semantic_matching > ≡ sample_job.txt
1  AI Engineer
2
3  Required Skills:
4  Python
5  Machine Learning
6  FastAPI
7  Docker
8
9  Responsibilities:
10 Develop AI recruitment systems.
11 Build scalable APIs using FastAPI.
12 Work on Machine Learning projects.
```

```
semantic_matching > ≡ sample_resume.txt
1  Software Engineer
2
3  Skills:
4  Python
5  FastAPI
6  Machine Learning
7  TensorFlow
8  Docker
9
10 Experience:
11 Built AI-powered recruitment systems using Python and Machine Learning.
12 Developed REST APIs with FastAPI.
```